

Chathura Dharmasiri

AI/ML Engineer – Intern

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SUMMARY

AI/ML-focused BSc (Hons) Computer Science undergraduate with hands-on experience building Retrieval-Augmented Generation (RAG) systems, agentic LLM applications with LangChain and LangGraph, and machine learning pipelines in a supervised research project on real-time Formula 1 overtake prediction. Completed Harvard's CS50 Introduction to AI with Python. Experienced in taking AI systems from prototype to deployment with FastAPI, Next.js and Docker, with a focus on grounded outputs, guardrails and evaluation. Eager to contribute to an AI engineering team through a focused internship.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, MATLAB

AI & LLMs: RAG, LangChain, LangGraph, Google Gemini API, Groq, Prompt Engineering, Vector Embeddings, Semantic Search, Guardrails, LLM Evaluation, LangSmith

ML & Data: XGBoost, Scikit-learn, Pandas, NumPy, Feature Engineering, Time-Series Analysis, FastF1 Telemetry

Vector & Databases: ChromaDB, PostgreSQL, ScyllaDB (Cassandra CQL)

Backend & Frontend: FastAPI, REST APIs, SSE Streaming, Next.js, React, Tailwind CSS

Tools & DevOps: Docker, Git, GitHub Actions, Cloudflare Pages, Vercel, Linux/Ubuntu

PROJECTS & RESEARCH

DocuMind AI – RAG Document Intelligence Platform

2026

Next.js, FastAPI, Gemini API, ChromaDB, PyMuPDF, PostgreSQL, Docker

[GitHub](#) | [Live Demo](#)

- Built an end-to-end **RAG** platform with **FastAPI** and **Next.js** where users upload multi-page PDFs and get grounded answers and automated summaries.
- Engineered an ingestion pipeline using PyMuPDF with OCR fallback, sliding-window chunking (500 tokens, 10% overlap) and Gemini embeddings, indexed into **ChromaDB** with per-document metadata filtering.
- Reduced hallucinations with constrained context-grounding prompts and source-snippet attribution so every answer links back to the retrieved passages.
- Hardened API security by removing IDOR risks with UUIDv4 document IDs, enforcing MIME/magic-byte file validation (25 MB limit) and isolating storage per client.
- Delivered a responsive UI with live Markdown rendering, source inspection and dark/light mode, deployed on **Cloudflare Pages** and **Docker**.

Inferring Battery State-of-Charge (SoC) and Predicting Overtaking Probability in Formula 1 Racing Using Machine Learning Under 2026

Aug 2026 – Ongoing

Power Unit Regulations

Final-Year Research Project (CSH4216) | Python, XGBoost, ScyllaDB, FastAPI, React 19, FastF1, Docker

[GitHub](#)

- Architected a real-time telemetry streaming and predictive analytics platform that ingests live F1 SignalR timing feeds and predicts overtake probability for the full 20-car grid.

- Trained and evaluated an **XGBoost** classifier on 10M+ rows of multi-round Grand Prix telemetry, handling severe class imbalance with `scale_pos_weight` and PR-AUC across 10 features (closing delta, speed, throttle, braking, DRS, RPM, tyre degradation, energy state).
- Modeled a physics-based battery State-of-Charge engine aligned with the FIA 2026 power unit rules (350 kW MGU-K), simulating four energy-harvesting modes to estimate battery advantage in wheel-to-wheel battles.
- Built a low-latency streaming pipeline on **ScyllaDB** and **AsyncIO** with lockless atomic state swaps (`os.replace`) for high-frequency ingestion without buffer drops.
- Developed a 10 Hz live telemetry dashboard in **React 19** and Tailwind CSS with head-to-head battle gauges, closing-speed and energy-differential meters.

SmartShop AI – Multimodal Shopping Agent

Ongoing

Next.js, FastAPI, LangChain, LangGraph, Gemini, PostgreSQL, Docker, Langfuse, GitHub Actions

GitHub

- Architected an end-to-end multimodal AI shopping assistant using **FastAPI**, **Next.js** and **LangGraph**, with real-time SSE streaming recommendations and image-based receipt/product search.
- Engineered a 3-tier, zero-cost guardrail pipeline (regex filters plus structured Gemini classifier) to block off-topic and prompt-injection queries before they reach the main LLM.
- Designed a database validation layer on **PostgreSQL (asyncpg)** that checks every recommended product’s price and stock to prevent hallucinated items.
- Built multi-turn agentic workflows with LangChain and LangGraph that persist dietary preferences, order history and budget constraints as personalized agent memory.
- Set up CI/CD and evaluation pipelines with **GitHub Actions** and **Langfuse**, benchmarking tool-calling accuracy on golden datasets under free-tier rate limits.

EDUCATION

University of Vavuniya	Sri Lanka
BSc (Hons) in Computer Science	Sept. 2023 – Present

CERTIFICATIONS

CS50’s Introduction to Artificial Intelligence with Python	Sep 2026
Harvard University	Credential
MATLAB Onramp	Sep 2026
MathWorks	Credential
Python Programming – Trainee Full Stack Developer	May 2024
University of Moratuwa	Credential ID: WgGERcT8FF Verify

SOFT SKILLS

Problem Solving Teamwork Adaptability Time Management Quick Learner Can-Do Attitude

REFERENCES

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